

## Written Web Design L-3

**1. What does CSS stand for in web design?**

- a) Computer Style Sheets
- b) Creative Style Solutions
- c) **Cascading Style Sheets**
- d) Centralized Style System

**2. Which of the following HTML tags is used to create a hyperlink?**

- a) <link>
- b) **<a>**
- c) <url>
- d) <href>

**3. Which programming language is primarily used for client-site scripting in web development?**

- a) Java
- b) Python
- c) **JavaScript**
- d) Ruby

**4. Which of the following is NOT a commonly used web design file format for images?**

- a) .jpg
- b) **.mp3**
- c) .png
- d) .gif

**5. What is the purpose of the “alt” attribute in HTML image tags?**

- a) It specifies the alignment of the image.
- b) It defines the size of the image
- c) **It provides alternative text for the images for accessibility.**
- d) It controls the animation of the image

**6. How can you include CSS in an HTML document?**

- a) Using the <style> tag within the <head> section?
- b) Using the <link> tag with a “stylesheet” relationship in the <head> section
- c) Using the style attribute in HTML elements
- d) **All of the above**

**7. Which CSS property is used to control the spacing between lines of text?**

- a) text-spacing
- b) line-spacing
- c) letter-spacing
- d) **line-height**

**8. What does the “float” property in CSS do?**

- 1. It Controls the alignment of text within an element.
- 2. **It removes an element from the normal flow of the document and positions it to the left or right of its containing element.**
- 3. It makes an element visible or hidden.
- 4. It rotates an elements clockwise or counterclockwise.

**9. What does the “z-index” property in CSS control?**

- a) It controls the size of an element.
- b) It controls the position of an element in the stacking order.**
- c) It controls the opacity of an element.
- d) It Controls the rotation of an element.

**10. Which HTML tag is used to define a table row?**

- 1. <tr>**
- 2. <td>
- 3. <th>
- 4. <table>

**11. Which keyword is used to declare a variable in javascript?**

- 1. Var**
- 2. Variable
- 3. V
- 4. Let

**12. Which operator is used for strict equality comparison in JavaScript?**

- a) ==
- b) !=
- c) ===**
- d) !==

**13. Which function is used to output data to the console in JavaScript?**

- a) Print()
- b) Log()
- c) Display()
- d) Console.log()**

**14. Which built-in method is used to convert a string to an integer in JavaScript?**

- a) parseInt()**
- b) toInteger()
- c) convertToInt()
- d) stringToInt()

**15. What does the method querySelector() do in JavaScript?**

- a) It selects multiple elements based on a class name.
- b) It selects the first element based on a CSS selector.**
- c) It selects an element based on its ID
- d) It selects all elements of a specific tag name.

**16. What are the three main components involved in a Client-Server setup?**

Answer: The three main components in a Client-Server setup are: Client, Server, and Network.

**17. What is significance of adhering to web standards in achieving cross browser compatibility?**

Answer: Adhering to web standards ensures that websites function consistently across different browsers and devices.

**18. How does an FTP client work?**

Answer: An FTP client connects to a server using FTP protocol to upload, download, and manage files.

**19. How can you create a responsive navigation bar using Bootstrap?**

Answer: Use Bootstrap's navbar component with classes like navbar, navbar-expand, and navbar-collapse for responsiveness.

**20. How can you create a carousel of images using Bootstrap?**

Answer: Use Bootstrap's carousel component with carousel, carousel-inner, carousel-item, and data-bs-ride attributes.

**21. What role does and database play in a CMS?**

Answer: A database stores and retrieves content and settings for a CMS, such as posts, users, and pages.

**22. How can Bootstrap be integrated into a project?**

Answer: Bootstrap can be integrated by linking its CDN in HTML or downloading and importing its CSS and JS files.

**23. What are three common features of a CMS?**

Answer: Three common CMS features: Content editing, User management, and Theme/plugin support.

**24. What is AJAX in jQuery?**

Answer: AJAX in jQuery allows asynchronous data loading from a server without refreshing the page.

**25. How does jQuery simplify DOM manipulation?**

Answer: jQuery simplifies DOM manipulation using concise syntax like \$(selector).action() for selecting and modifying elements.

**26. How does a CMS enable collaboration among multiple users?**

Answer: A CMS enables collaboration by allowing multiple users with different roles to create, edit, and manage content.

**27. What is function in Excel? Write an example?**

Answer: A function in Excel performs calculations. Example: =SUM(A1:A5) adds values in cells A1 to A5.

**28. What is the purpose of slide transitions in PowerPoint?**

Answer: Slide transitions enhance presentations by adding visual effects when moving between slides.

**29. How can you add animations to objects in PowerPoint?**

Answer: Select the object, go to the Animations tab, and choose an animation effect from the gallery.

**30. How can you spell check a document in Microsoft Word?**

Answer: Go to Review > Spelling & Grammar to check and correct spelling in Word.

**31. What are the Fundamental Steps of Image Processing?**

Answer: 1. Image Acquisition 2. Image Enhancement 3. Image Restoration 4. Color Image Processing 5. Wavelets and Multi-Resolution Processing 6. Compression 7. Morphological Processing 8. Segmentation 9. Representation and Description 10. Object recognition 11. Knowledge Base

**32. Differentiate between WordPress.com and WordPress.org.**

Answer: WordPress.com is a hosted platform with limited customization, ideal for beginners. WordPress.org requires self-hosting, gives full control, and is better for businesses and full customization.

**33. Why do we use Jumbotron in Bootstrap?**

Ans: Jumbotrons are large, responsive containers used to:

- Capture attention
- Showcase important content
- Provide a call to action

**34. What is a Bootstrap Grid System?**

Answer: Bootstrap's 12-column grid system used to create responsive layouts for web pages that work across screen sizes.

**35. What is contextual classes of table in Bootstrap?**

Answer: Bootstrap offers a set of contextual classes that can be applied to tables to visually indicate the state or condition of the table or its individual rows and cells. These classes help to provide visual cues to users, making it easier to understand the content and identify important information.

**36. What is the purpose of the nth-child() pseudo-class in CSS3?**

Answer: The nth-child() CSS pseudo-class selector matches the element based on the index of the elements in the child list of its parent.

The selected position is determined by an integer, keyword, or expression located inside the pseudo-class's parentheses.

**37. What is a pseudo-element selector?**

Answer: A CSS pseudo-element is used to style specified parts of an elements. For example, it can be used to: style the first letter, or line, of an elements.

**38. How to use class selector?**

Answer: Class selector is used to select all elements which belong to a particular class attribute. In order to select the elements with a particular class, use the period (.) character specifying the class name it will match the HTML element based on the contents of their class attribute.

**39. What is the default positioning scheme in CSS3?**

Answer: Static is the default position for HTML elements.

**40. Which property is used to add shadows to text in CSS3?**

Answer: The CSS text shadow property is used to add a shadow to the text. Multiple property parameters can be specified, separated by a space.

**41. What is HTML Elements?**

Answer: HTML elements are the building blocks of HTML documents. They represent different parts of a web page, such as text, image, links, heading, paragraphs, and more. Each element is defined by a start tag, content, and end tag.

**42. What is and IDE?**

Answer: An integrated development environment is a software application that helps programmers develop software code efficiently.

**43. What is a rowspan in HTML?**

Answer: The rowspan attribute specifies the number of rows a cell should span.

**44. What are the basic SVG tags?**

Answer: Inside the <svg> tag, we can use various elements to draw shapes:

- <circle> for circles
- <rect> for rectangles
- <line> for straight lines

<path> for complex shapes and paths  
<text> for text

**45. What syntax we use to include a video on the webpage for HTML video element?**

Answer: To include a video on your webpage use the following syntax:

```
<video controls>  
    <source src = "movie.mp4" type = "video/mp4" />  
</video>
```

**46. What are the plugins in HTML?**

Answer: HTML plugins are pieces of code that we can insert into your HTML files to add features or effects for example you can use HTML plugins to create sliders, animations, charts, calendars, pop-ups and more.

**47. How do you handle conditional statements in JavaScript? Explain the if, else, and switch statements.**

Answer: Conditional statements are used to execute different code blocks based on different conditions.

“if” statement: Executes a block of code if a specified condition is true.

“else” statement: Executes a block of code if the same condition is false.

“Switch” statement: Evaluates an expression and executes a block of code depending on matching cases.

**48. How do select and manipulate HTML elements using JavaScript?**

Answer: HTML elements can be selected and manipulated using methods like getElementById(), getElementsByClassName(), querySelector(), and manipulating their properties like innerHTML, style etc.

**49. How do you handle user input validation in JavaScript?**

Answer: User input validation involves checking user-provided data against expected criteria to ensure it meets the required format, length or range.

**50. What are the benefits of using a headless CMS with a static website?**

The benefits of using a headless CMS with a static website include separating content management from presentation, enabling content reuse across multiple channels and providing a user friendly interface for non-technical users to manage content.

**51. Which tag is used to create a dropdown list in a form?**

- a) <input>      **b) <select>**      c) <option>      d) <list>

**52. Which of the following is an example of an HTML entity?**

- a) &copy;**      b) <p>      c) #header      d) color:red;

**53. Which preprocessor allows nesting and variables in CSS?**

- a) SASS**      b) Bootstrap      c) Tailwind CSS      d) jQuery

**54. What is the main difference between JPEG and PNG formats?**

- a) JPEG supports transparency, while PNG does not.

**b) PNG supports transparency, while JPEG does not.**

- c) JPEG is a lossless format, while PNG is lossy.

- d) PNG is always smaller in file size than JPEG.

**55. What is the benefit of using smart objects when designing UI?**

- a) They automatically write JavaScript code for the UI.

- b) They convert raster images into pure vector graphics.

**c) They allow reusable, editable components.**

- d) They reduce the file size of all design assets automatically.

**56. Which method is used to select an element by ID in the DOM?**

- a) document.querySelectorAll()

**b) document.getElementById()**

- c) document.getElementsByClassName()

- d) document.getElementsByTagName()

**57. How do you define a JavaScript function?**

**a) function myFunction() { // code }**

- b) def myFunction() { // code }

- c) create function myFunction { // code }

- d) function: myFunction() => { // code }

**58. Which function checks if a value is a number in JavaScript?**

- a) isNumber()      b) Number()      **c) isNaN()**      d) typeof()

**59. What is debugging in JavaScript?**

a) The process of writing JavaScript code for a web page.

**b) The process of finding and fixing errors or bugs in JavaScript code.**

c) The process of compiling JavaScript into machine language.

d) The process of styling a web page using JavaScript.

**60. Is it possible to SEO a WordPress website to show it on Google's first page?**

**Ans.** Yes. by using SEO plugins, quality content, keyword, and fast loading designs.

**61. What is the difference between a post and a page in WordPress?**

**Ans. Post:** Time based content for blogs.

**Page:** Static content like "About" or "Contact"

**62. What is a container in Bootstrap and what is it used for?**

**Ans.** A container centers and pads content in the layout using **.container** or **.container-fluid**.

**63. What is the Bootstrap Grid System and what is it used for?**

**Ans.** It's a 12-column layout system using **.container**, **.row** and **.col** to build responsive designs.

**64. Which class is used to display informational messages in Bootstrap?**

**Ans.** **.alert-info**, **.text-info** or **.bg-info**

The classes **.text-info** or **.bg-info** are used to indicate informational content with a light blue color.

**65. What are the CSS methods?**

**Ans.** Inline CSS, Internal CSS and External CSS

**66. What is media query in CSS?**

**Ans.** A media query applies styles based on screen size or device type for responsive design.

Example: `@media(max-width:600px){`

```
    body{  
        font-size:14px;  
    }  
}
```

**67. What is pseudo-classes in CSS?.**

**Ans.** A keyword added to a selector to define a special state(e.g., **:hover**, **:focus**)

**68. How do you write a pseudo- element?**

**Ans.** Use two colons. Example: p::first-letter

**69. What is semantic HTML, and why is it important?**

**Ans.** Semantic HTML uses meaningful tags like <header>, <article>, <section>, and <footer> to clearly describe the content's purpose. It improves accessibility, SEO and code readability, helping both developers and search engines understand the structure.

**70. What does the 'alt' attribute in the <img> tag do, and why is it important?**

**Ans.** It provides alternative text for images, improving accessibility and SEO.

**71. What is definition list in HTML?**

**Ans.** A list that defines terms using <dl>, <dt>(term) and <dd>(definition).

**72. What are the key components that make up the infrastructure of the Internet?**

**Ans.** Servers, clients, routers, IP addresses, DNS and protocols like HTTP/HTTPS.

**73. What are variables in Javascript and how do you declare them?**

**Ans.** Variables store data. Use var, let or const to declare them.

**74. What are arrays and objects in JavaScript? How do you access their elements/properties?**

**Ans.** Array: List of items, accessed by index, arr[0]

Object: key-value pairs, accessed by key.obj.name

**75. What is the Document Object Model (DOM) in JavaScript?**

**Ans.** A tree-like structure representing the HTML page, allowing scripts to manipulate elements dynamically.

**76. How do you handle errors in JavaScript?**

**Ans.** Use try\_\_catch blocks to catch and manage errors.

**77. What are Bootstrap JavaScript components?**

**Ans.** Interactive features like modals, dropdowns, tooltips and carousels powered by Bootstrap's JS.

**78. What are the key components of Bootstrap?**

**Ans.** Grid system, utilities, layout, content (typography/images), components (buttons, modals), and JavaScript plugins.



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1. How do you ensure accessibility compliance when integrating JavaScript libraries into web applications?
  - A) By ignoring accessibility guidelines
  - B) By following web accessibility standards, using ARIA roles, and testing with screen readers and other assistive technologies**
  - C) By using only plain HTML
  - D) By minimizing the use of JavaScript
2. Which image format supports transparency and is commonly used for web graphics and images with transparent backgrounds?
  - A) PSD
  - B) PNG**
  - C) JPG
  - D) GIF
3. What is the function of the HTML style attribute?
  - A. It is used to add style to an HTML element**
  - B. It is used to uniquely identify some specific styles of some element
  - C. Both a & b
  - D. None of the above
4. What tag is used to render an image on a webpage?
  - A. img**
  - B. src
  - C. image
  - D. none of them
5. The most basic part of any HTML page is?
  - A. ASCII Text**
  - B. Binary Text
  - C. Text
  - D. None of the above
6. How many characters can be written in 1KB?
  - A. 1048
  - B. 1024**
  - C. 2048
  - D. None of them
7. Explain the difference between the ---- and ---- operators in JavaScript.
  - A) ---- compares values; ---- compares values and types**
  - B) ---- compares values and types; ---- compares only values
  - C) ---- compares objects;---- compares arrays
  - D) ---- compares strings;---- compares numbers
8. What are logical operators?
  - A) +, -, , /
  - B) &&, ||, !**
  - C) =, !=, ==, !==
  - D) if, else switch
9. How do you use the break statement in loops and switches in JavaScript?

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- A) To pause execution
  - B) To terminate a loop or switch**
  - C) To skip an iteration
  - D) To execution
10. How do you display output in JavaScript?
- A) Using print()
  - B) Using alert()
  - C) Using console.log()**
  - D) Using document.write()
11. What are single-line and multi-line comments in JavaScript? How do you comment out code?
- A) // for single-line, /\* for multi-line**
  - B) <!--> for single-line, <!--> for multi-line
  - C) # for single line, “””” “””” for Multi-line
  - D) // for single line, // for multi line
12. Select the basic data types in JavaScript.
- A) Number, String , Boolean, Object, Array
  - B) Number, String , Boolean, Undefined, Null, Object**
  - C) Integer, Float, String, Boolean, Object
  - D) Char, Float, String, Boolean, Array
13. How do you check the data type of a variable in JavaScript?
- A) Type of variable**
  - B) instance of variable
  - C) variable. type
  - D) check Type(variable)
14. What are arithmetic operators in JavaScript? Provide examples
- A) +,-,\*,/**
  - B) +, -, , /, && ,
  - C) ++, --, +=, -=
  - D) ==, ===, !=, !==
15. Which keyword is used to declare a variable in JavaScript?
- A) var**
  - B) variable
  - C) v
  - D) let
16. Which operator is used for strict equality comparison in JavaScript?
- A) ==
  - B) !=
  - C) ===**
  - D) !==
17. Which function is used to output data to the console in JavaScript?
- A) print()
  - B) log()

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- C) display()
- D) console.log()**

18. Which built-in method is used to convert a string to an integer in JavaScript?
- A) parseInt()**
  - B) tointeger()
  - C) convert ToInt()
  - D) stringToInt()
19. What is the purpose of Tailwind CSS and how does it differ from other CSS frameworks like Bootstrap?
- A) Tailwind CSS is a JavaScript framework
  - B) Tailwind CSS is a utility-first CSS framework that provides low-level utility-----, while Bootstrap provides pre-designed components**
  - C) Tailwind CSS is used for server-side scripting
  - D) Tailwind CSS is a database management tool
20. What is the basic difference between JavaScript and Java?
- A. Functions are considered as fields
  - B. Functions are values, and there is no hard distinction between methods and fields**
  - C. Variables are specific
  - D. There is no difference
21. Which statement is used to execute a block of code multiple times in JavaScript?
- A. for**
  - B. if
  - C. return
  - D. break
22. Why event handlers are needed in JS?
- A. Allows JavaScript code to alter the behavior of windows**
  - B. Adds inner HTML page to the code
  - C. Change the server location
  - D. Performs handling of exceptions and occurrences
23. What is the purpose of JavaScript in web development?
- A. To structure a web page
  - B. To style web page
  - C. To add interactivity and dynamic content to web pages**
  - D. To store data on the server
24. In JavaScript, which of the following is a valid variable name?
- A. 2names
  - B. \$name**
  - C. -name
  - D. @name
25. Which of the following class is used to create a responsive navigation bar in Bootstrap?
- A. navbar-fixed
  - B. navbar-expand**

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- C. navbar-light
- D. navbar-collapse

26. What is the purpose of the card class in Bootstrap?

- A. To create interactive buttons.
- B. To apply borders to an element.
- C. To create a flexible content container with optional headers and footers.**
- D. To align text within a grid.

27. How can we write comments in CSS?

- A. //**
- B. //
- C. #
- D. All of this

28. The CSS property is used to control the transparency of an element?

- A. Opacity.**
- B. Visibility
- C. filter
- D. None of them

29. Which of the following CSS property is used for controlling the layout?

- A. display**
- B. color
- C. text
- D. None of them

30. Which of the following components of the CSS box model are transparent?

- A. padding
- B. margin
- C. both a & b**
- D. content

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31. How many types of layout are available in Bootstrap?

There are two major layouts for Bootstrap:

- A) Fluid Layout- This layout is necessary for creating an app that is 100 % wider and covers all the screen width.
- B) Fixed Layout- It is used only for a standard screen (940px). Both layouts can be used for creating a responsive design.

32. Write down the boilerplate of HTML?

```
<!DOCTYPE html>
<html>
  <head>
    <meta charact=" UTF-8"/>
    <title>Document Title</title>
  </head>
  <body>
  </body>
</html>
```

33. Why we use CSS box-sizing properly?

Box sizing property remove Border & Padding from height & width property in CSS

34. What is the difference between margin & padding?

- Margin is used to create space around elements and padding is used to create space around elements inside the border.
- We can set the margin property to auto but we cannot set the padding property to auto.
- In Margin property we can allow negative or float number but in padding we cannot allow negative values.

35. What is JSON & give an example JSON?

- JSON is a standard data interchange format that's used to store simple data structures and objects. It's a lightweight, language-independent format that's easy to learn and troubleshoot
- Var student = {name: "Rubel", city: "Dhaka" }

36. What are the timing method in JavaScript & Why we use it?

There are two timing method in JavaScript. Setinterval & Settimeout. These method is used for creating task repeatedly. Setinterval execute a task repeatedly within given time period.

Settimeout execute a task one time within given time period.

37. What is the difference between DOM & BOM?

- DOM stands for Document Object Model. DOM is a programming interface for HTML and XML documents.
- Browser Object Model is known as BOM. It allows users to interact with the browser. A browser's initial object is a window. As a result, you may call all of the window's functions directly or by referencing the window.

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38. What are the difference between HTML TAG, ELEMENT & ATTRIBUTE?

- HTML Tag is a syntax that is used to print data & define data print.
- HTML Element contain starting tag & ending tag with content.
- HTML Attribute provide additional information about HTML Tag

39. Describe different type HTML Element?

- There are two type of HTML Element,
- Block Label Element & Inline Element.
- Block Label element hold the whole area.
- Inline element hold only contain area.

40. What is function in JavaScript? Give an example?

Function is a block of code that has a particular job. Function can not work itself. When we call a function then it will work. Function can reuse.

```
function hello()
{
    Document.write("Hello World.");
}
```

41. What are Array in JavaScript? Give an example?

- Array is a special variable in JavaScript. Array can store multiple data. In JavaScript it is possible to store different type of data in an Array.
- Var data = [10, 20, "Hello World.", 20.34, true, false];

42. Create a JSON Array for that contain four students name, email, mobile & address.

```
Var students = [
    {name: "Arif", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
    {name: " Arif ", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
    {name: " Arif ", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
    {name: " Arif ", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
]
```

43. Describe different type of repeated statement in JavaScript?

- In JavaScript there are for, while, do while, foreach, forin repeated statement.
- For & while syntax start works when condition is true. Do while syntax one time works after check condition, if the condition is true then it works again.
- Foreach & Forin syntax works for Array.

44. Describe different type of conditional statements syntax in JavaScript?

- There are four conditional statement syntax in JavaScript. They are if, if else, if else if & switch.
- If syntax work if the condition is true.
- If else syntax work if the condition is true & if not true then it show another result.

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- If else if syntax work for multiple condition.
- Switch syntax work for property value. Inside switch syntax there are lot of case define. Which property value matches with case then that particular case will start execute statement & it stop after break syntax.

45. Create a registration form using only HTML. Form contain user name, email, mobile & address?

```
<form>
  <div>
    <label>User Name</label>
    <input type="text" name="name"/>
  </div>
  <div>
    <label>Email</label>
    <input type="email" name="email"/>
  </div>
  <div>
    <label>Mobile</label>
    <input type="number" name="mobile"/>
  </div>
  <div>
    <label>Address</label>
    <textarea name="address"></textarea>
  </div>
</form>
```

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46. What is the difference between inner Text and inner HTML in JavaScript?

- **innerText**: Gets or sets the text content of an element, excluding hidden elements. It respects CSS styling (like `display: none` and `visibility: hidden`).
- **innerHTML**: Gets or sets the HTML content inside an element, including tags. It parses the string as HTML, allowing for rendering of nested elements.

47. How does an interactive website differ from static and namic website?

- **Static Website**: Contains fixed content (HTML/CSS). It doesn't change without manual updates and has no user interaction (e.g., simple portfolio sites).
- **Dynamic Website**: Displays dynamic content using server-side scripting (e.g., PHP, Node.js). It fetches data from databases, allowing content updates without modifying the code (e.g., blogs, news sites).
- **Interactive Website**: Goes beyond dynamic content by engaging users through real-time interactions, animations, and user-driven experiences. It uses client-side scripting (JavaScript) for features like forms, chatbots, and dynamic navigation (e.g., social media platforms, web apps).

48. Which code editor do you prefer for web design and development projects, and why?

I prefer **Visual Studio Code (VS Code)** because it's fast, highly customizable, and has great extensions for HTML, CSS, and JavaScript. It also offers built-in Git integration, smart autocompletion, and a Live Server for real-time previews, making web development efficient and seamless.

49. Can you explain the basic purpose of FTP (File Transfer Protocol ) software in web development?

FTP (File Transfer Protocol) software is used in web development to **upload, download, and manage files** on a web server. It allows developers to transfer website files (like HTML, CSS, images) from their local computer to the server, making the site accessible online. It's also used for updating and maintaining websites.

50. How do you create HTML login form with submit button?

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Form</title>
</head>
<body>
  <h2>Login Form</h2>
  <form action="login.php" method="post">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required><br><br>

    <button type="submit">Login</button>
  </form>
</body>
```



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</html>

51. When should I consider using SVG image?

Use **SVG** when you need scalable, high-quality graphics that are lightweight, editable with CSS/JavaScript, and accessible for SEO. Ideal for logos, icons, and simple illustrations.

52. What is an ID selector in CSS?

An ID selector in CSS is used to style a specific element with a unique identifier. It's defined using a # followed by the element's ID. Example:

```
#header {  
    color: blue;  
    font-size: 20px;  
}
```

53. How do you apply multiple CSS classes to a single HTML element?

```
<div class="class1 class2 class3">  
    Content here  
</div>
```

54. How do you create functions in JavaScript?

**Function Declaration:**

```
function greet(name) {  
    return "Hello, " + name;  
}
```

**Arrow Function:**

```
const greet = (name) => {  
    return "Hello, " + name;  
}
```

55. How do you create an ordered list in HTML?

To create an ordered list in HTML, use the <ol> (ordered list) tag, with each list item wrapped in an <li> (list item) tag:

```
<ol>  
    <li>First item</li>  
    <li>Second item</li>  
    <li>Third item</li>  
</ol>
```

56. What are arrays and objects in JavaScript?

- **Arrays:** Arrays in JavaScript are used to store multiple values in a single variable. They are ordered collections and can hold any type of data.
- **Objects:** Objects are collections of key-value pairs, where each key is a string (or symbol) and the value can be any data type.

57. How do you access arrays and objects and their elements properties?

- **Array:** Access elements using an index, e.g., `fruits[0]`.
- **Object:** Access properties using dot or bracket notation, e.g., `person.name` or `person["age"]`.

58. Can you explain the difference between for loop, while loop, and do-while loop?

## Web Design Level 3

- **\*\*for loop\*\***: Used when the number of iterations is known in advance.
- **\*\*while loop\*\***: Runs as long as the condition is true, checked before each iteration.
- **\*\*do-while loop\*\***: Runs at least once, condition checked after each iteration.

59. How do you handle conditional statements in JavaScript?

Use if to check a condition, else for the alternative, and else if for multiple conditions. The switch statement checks different values.

```
if (x > 10) { console.log("x > 10"); }  
else if (x === 10) { console.log("x = 10"); }  
else { console.log("x < 10"); }
```

60. What is the Document Object Model (DOM) in JavaScript?

The Document Object Model (DOM) is a programming interface for web documents. It represents the page as a tree structure, where each element, attribute, and text is a node. JavaScript can manipulate the DOM to change the content, structure, and style of a webpage dynamically.

For example, you can use JavaScript to change the text inside a <div>:

```
document.getElementById("myDiv").innerHTML = "New content!";
```

# Web Design Level 3

1. How do you ensure accessibility compliance when integrating JavaScript libraries into web applications?
2. Which image format supports transparency and is commonly used for web graphics and images with transparent backgrounds?
3. Explain the difference between == and === operators in JavaScript.
4. What are logical operators?
5. How do you use the break statement in loops and switches in JavaScript?
6. How do you display output in JavaScript?
7. What are single-line and multi-line comments in JavaScript? How do you comment out code?
8. Select the basic data types in JavaScript.
9. How do you check the data type of a variable in JavaScript.
10. What are arithmetic operators in JavaScript? Provide examples.
11. Which keyword is used to declare a variable in JavaScript?
12. Which operator is used for strict equality comparison in JavaScript?
13. Which function is used to output data to the console in JavaScript?
14. Which built-in method is used to convert a string to an integer in JavaScript?
15. What is the purpose of Tailwind CSS and how does it differ from other frameworks like Bootstrap?
16. What is the difference between innerText and innerHTML in JavaScript?
17. How does an interactive website differ from static and dynamic websites?
18. Which code editor do you prefer for web design and development projects, and why?
19. Can you explain the basic purpose of FTP(File transfer Protocol) software in web development?
20. How do you create an HTML login form with a submit button?
21. When should I consider using SVG images?
22. What is an ID selector in CSS?
23. How do you create functions in JavaScript?
24. How do you create an ordered list in HTML?
25. What are arrays and objects in JavaScript?